

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

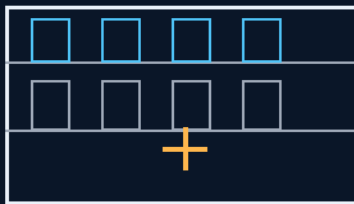
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 204

THE SHIP'S PHARMACY

iodine and antibiotics first
dry-store beats wet-stock — carry the recipe



a century ship's formulary, grouped by intent

ANTISEPSIS FIRST · CLASS DIVERSITY · DRY STORE
THE PHARMACY IS A RECIPE, NOT A WAREHOUSE

Medical — v1.0 in force

GP-IM-204

Revision v1.0 — 24 August 2026

205 of 290

VETTED BATCH 26 — PASS AS DOCTRINE · MEDICAL VALIDATION SEPARATE

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 204

PHASE 204: THE SHIP'S PHARMACY — THE MUST-HAVES LIST — STATUS: CURRENT — PASS AS AN INVENTORY/DESIGN DOCTRINE; MEDICAL VALIDATION SEPARATE FROM PHYSICS. The pharma queue closes with the must-haves list — the inventory a century ship's pharmacy must carry and make, with the Author's ordering standing: iodine and antibiotics first. The list, grouped by intent: antiseptics first — liquid Betadine (povidone-iodine), Condyl's crystals (potassium permanganate, the dry-store alternative, stable oxidizer with expedition heritage), sulphur powder, alcohol; antibiotics — amoxicillin, penicillin, and two others from different classes (his diversity clause, the resistance answer); pain and fever — codeine, paracetamol, aspirin or ibuprofen, three mechanisms; respiratory and emergency — ephedrine (his note: easy), salbutamol, epinephrine; steroids — cortisone injectable and topical; hormonal and urological — estrogen, tamsulosin; theatre — injectable sedative and surgical anesthesia; topical — antifungal cream and powder; field chemistry — snap packs with prill and water, instant cold packs by endothermic dissolution, stored dry, hydrated when needed. The doctrine read from the list: dry-store beats wet-stock; class diversity is the resistance strategy; synthesis-easy markers noted for the methodologies sitting. His amendment answers the same-class flag at supply-chain level: amoxicillin is fully synthetic, the other is not — one may be more easily made; 'humungus fungus may exist on some planets.' The vet passed it as inventory/design doctrine, with shelf life, synthesis, sterility, dosing and clinical protocols fenced to medical validation — not physics vetting.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-204, v1.0 — 24 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the pharmacy law as seated (contents line, the bodies — the list by intent, the doctrine read — the rulings, the audit and provenance, verbatim) — §2 why dry-store and class-spread win (graded) — §3 the system on the shelf — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 204: **The Ship's Pharmacy — The Must-Haves List**. Iodine and antibiotics first; Betadine/Condy's crystals/sulphur/alcohol antiseptics; amoxicillin + penicillin + two diverse classes; analgesic trio; ephedrine/Ventolin/epinephrine; cortisone; estrogen/tamsulosin; theatre set; antifungals; prill snap-packs. Methodologies sitting to follow.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Coverage of the Killers and the Daily — stocked dry where possible, synthesized where sensible '

Medical Baseline: Pharmacy Inventory & Synthesis Doctrine — Century-Voyage Formulary

Live instruction, 31 July 2026. The pharma queue closes with the must-haves list — the inventory a century ship's pharmacy must carry and make. The Author's ordering stands: iodine and antibiotics first. Methodologies — how the ship synthesizes rather than stockpiles — are the next pharma sitting; this phase is the list and its doctrine.

THE LIST — AS GIVEN, GROUPED BY INTENT (VERBATIM)

- **Antisepsis first:** liquid Betadine (povidone-iodine); Condy's crystals (potassium permanganate — his dry-store alternative, stable oxidizer with expedition heritage); sulphur powder; alcohol.
- **Antibiotics:** amoxicillin, penicillin, and two others from different classes — his diversity clause, the resistance answer.
- **Pain and fever:** codeine, paracetamol, aspirin or ibuprofen — three mechanisms, combinable cover.
- **Respiratory and emergency:** ephedrine (his note: easy), Ventolin ingredients (salbutamol), epinephrine.
- **Steroids:** cortisone, injectable and topical.
- **Hormonal and urological:** estrogen; tamsulosin.
- **Theatre:** injectable sedative and surgical anesthesia.
- **Topical:** antifungal cream and powder.
- **Field chemistry:** snap packs with prill and water — gel likely to fill packs later on; instant cold packs by endothermic dissolution, stored dry, hydrated when needed.

THE DOCTRINE READ FROM THE LIST (KIMI) (VERBATIM)

- **Dry-store beats wet-stock:** Condy's crystals for liquid, prill for gel — mass and shelf-life physics applied to medicine: carry the stable dry form, make the working form on demand.
- **Class diversity is the resistance strategy:** two antibiotics from different classes means no single

resistant bug clears the shelf.

- **Synthesis-easy markers noted by the Author:** ephedrine tagged '(easy)' — the methodologies sitting will sort the list into synthesizable versus stockpile-deep.

PHYSICS NOTE — THE AUDIT (KIMI: STRUCTURE AFFIRMED — THE FORMULARY REVIEW IS A PHARMACIST'S BENCH, SO FLAGGED) (VERBATIM)

The list's architecture is affirmed: antiseptics before antibiotics is triage logic (the wound you clean needs fewer drugs than the wound you don't); the analgesic trio covers three mechanisms; the dry-store instinct is correct physical chemistry — potassium permanganate keeps for decades dry where iodine solutions degrade, and prill-plus-water cold packs are endothermic dissolution doing injury duty, with the gel-fill-later note preserving payload mass exactly as his supply doctrine does everywhere else. And a note the file should not miss: prill returns — the fertilizer his father taught him the chemistry of at eight years old, back aboard as medicine's ice. Honest flags, all standing: (1) amoxicillin and penicillin are the SAME class — beta-lactams — so his 'two others from different classes' clause carries the whole diversity load; the usual reserve classes (macrolide, tetracycline, fluoroquinolone families) are named here as audit commentary, and the final picks are a pharmacist's bench, flagged, never prescribed by the audit. (2) Shelf-life is the century problem — epinephrine notoriously degrades, antibiotics lose potency; the list therefore splits in the methodologies sitting into stockpile-rotation versus synthesizable-on-demand, and that split is the real design question, flagged for that sitting. (3) Controlled substances — codeine, ephedrine, sedatives, anesthetics — need a custody and dispensing doctrine in Lex; flagged for the law volume, not designed here. (4) The audit is not a pharmacist: this phase records inventory doctrine, not dosing, not protocols, not prescribing — those belong to the flight surgeon the file has always assumed, and the no-invention rule holds hardest exactly here. (5) Condy's crystals identified from his phonetic 'condees' — potassium permanganate, antiseptic and water-treatment double duty; his spelling stands in the quotes, the chemistry name in the ledger.

"amomxacillin is fully synthetic, the other is not, one may be more easily made, its why i put both, humungus fungus may exist on some planets"

The audit flagged amoxicillin and penicillin as one chemical class (beta-lactams); the Author's reply is that the pairing was never about chemical-class diversity — it is production-route diversity. One drug you make by chemistry on a bench; the other you grow by fermentation in a vat. Two molecules, two supply lines: if the fermentation line dies, the synthesis line still delivers a beta-lactam; if the synthesis precursors run out, the vat still grows one. That is redundancy in the route, not just the molecule, and it stands. One honest nuance, per the standing doctrine, recorded and never edited: industrially, amoxicillin today is semi-synthetic — it is built from 6-aminopenicillanic acid (6-APA), which is itself classically split out of fermentation-derived penicillin. Total chemical synthesis of 6-APA exists in principle but is not the industrial route; so as of Earth-present, both routes still trace partly back to the same vat.

The Author's "fully synthetic" is kept verbatim as his design intent, and the pharmacy methodologies sitting he has already named gets the bench question: whether the ship carries a total-synthesis line for 6-APA (closing the loop so the two drugs truly have independent supply chains), a fermentation line, or both — the pairing logic is affirmed either way, because on a century voyage even two routes that share an ancestor are still two routes. The "humungus fungus" clause is affirmed outright and seated as planetary bioprospecting doctrine: penicillin itself came from a mold — *Penicillium*, Fleming, 1928 — so a planet with fungal life is a potential antibiotic library no inventory can match. Honest flag beside it, per doctrine: any alien-organism culturing goes through the exo-biology survey and quarantine doctrines first — you do not brew an alien mold next to the crew air supply until it has been through survey, and the first contact with alien biochemistry is a sealed-lab event, not a galley event. The flag is the safety rail; the doctrine stands.

ORIGIN OF THOUGHT (VERBATIM)

Archer Quinn, 31 July 2026, verbatim:

"ok iodine and anti biotics first"

"amoxicillin, penicillin, and two others from different classes a liquid for of betadine , probably condees crystals as a dry store alternative, sulphur powder, codine, paracetamol, aspirin or ibuprofen, snap packs with prill and water gel likely fill packs later on ,estrogen, tamsulosin, ephedrine (easy), Ventolin ingredients, epinephrene, cortizone injectable and topical, injecable sedative, and surgical anesthesia, alcohol, anti fungal topical cream/powder"

Why it matters, in plain English: a century ship's pharmacy cannot be a warehouse — it must be a recipe. This list reads like one: dry crystals instead of bottles, classes spread so resistance can't clear the shelf, cold packs that weigh nothing until they're needed, and an honest '(easy)' beside the first drug the ship will learn to make. The voyage's medicine chest just got its first page — and it starts, correctly, with soap's stronger cousins and the two antibiotics everyone has heard of, before it ever gets fancy.

Stock the Ship's Pharmacy: the must-haves list, iodine and antibiotics first — a century-voyage formulary giving coverage of the killers and the daily; stocked dry where possible, synthesized where sensible. — INSTRUCTION RECONSTRUCTION (NOT VERBATIM — the original order was not preserved; reconstructed from the seated body, 10 August 2026, per sitting 147).

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

DRY-STORE BEATS WET-STOCK, AFFIRMED AS MASS AND SHELF-LIFE PHYSICS: Condy's crystals for liquid, prill for gel — carry the stable dry form, make the working form on demand. A century ship cannot carry a century of bottles; it carries the recipe and the stable precursors.

CLASS DIVERSITY IS THE RESISTANCE STRATEGY, AFFIRMED: two antibiotics from different classes means no single resistant bug clears the shelf; his amendment sharpens it at the supply-chain level — one fully synthetic, one not, so manufacture-aboard stays possible when one path fails.

THE SEPARATION OF DOCTRINES IS CORRECT, AFFIRMED AS THE VET READ IT: dosing and treatment protocols are kept out of this phase and assigned to the flight surgeon; the phase distinguishes stocked materials from methodologies still to be developed — the presence of a named medication is not a claim the ship can already manufacture it.

§3 — THE SYSTEM ON THE SHELF

- **Antisepsis first:** liquid Betadine; Condyl's crystals (dry-store alternative); sulphur powder; alcohol — the wound you clean needs fewer drugs than the wound you don't.
- **Antibiotics:** amoxicillin, penicillin, and two others from different classes — diversity against resistance, synthesis paths against supply failure.
- **Pain and fever:** codeine, paracetamol, aspirin or ibuprofen — three mechanisms, combinable cover.
- **Respiratory and emergency:** ephedrine (tagged 'easy' for the synthesis sitting), salbutamol, epinephrine.
- **The rest of the shelf:** cortisone injectable and topical; estrogen; tamsulosin; injectable sedative and surgical anesthesia; antifungal cream and powder.
- **Field chemistry:** snap packs — prill and water, gel to fill packs later; instant cold packs by endothermic dissolution, stored dry, hydrated when needed.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The dry-store doctrine is the file's mass law in medical dress: carry cycles and recipes, not consumables — the same law as the laundry (Phase 191) and the breathing set's layered redundancy (Phase 199).
- The pharmacy joins the medical book's formulary thread; the methodologies sitting — synthesize-aboard versus stockpile-deep — is named in the body as the list's next pass.
- The Author's ordering — iodine and antibiotics first — is triage logic seated as inventory law.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Stock the shelf in the Author's order: antiseptics first, then the antibiotic set across classes, then the mechanisms list.
- 2. Dry-stock everything that stores dry: Condy's crystals, prill packs, powders — hydrate on demand.
- 3. Tag the synthesis-easy markers (ephedrine first) for the methodologies sitting — synthesize-aboard versus stockpile-deep.
- 4. Seat the custody chain: controlled-drug custody delegated to the appropriate medical/legal system, dosing to the flight surgeon.
- 5. Bench the shelf-life ledger: rotation dates, stability per item, cold-chain exceptions — a pharmacist's review, as the vet fenced it.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 199 (the breathing set):** the layered-redundancy doctrine's medical counterpart — never one of anything the crew dies without.
- **Phase 191 (the wringer line):** the carry-cycles-not-consumables mass law, here applied to medicine.
- **The medical book:** the formulary and methodologies thread; the flight surgeon owns dosing and protocols.

§8 — DO-NOT-BUILD

- Do NOT seat dosing or treatment protocols in this phase — they belong to the flight surgeon; the separation is law.
- Do NOT wet-stock what stores dry — bottles are mass and shelf-life spent; crystals and prill are the doctrine.
- Do NOT single-class the antibiotic shelf — diversity is the resistance strategy; his synthesis amendment stands with it.
- Do NOT read a named medication as a manufacturing capability — stocked versus methodology is the phase's own distinction, fenced by the vet.

§9 — OWED

OWED: the pharmacist's bench the vet fenced — shelf life per item, synthesis routes for the tagged easies, sterility chain, the methodologies sitting's sort into synthesize-aboard versus stockpile-deep; controlled-drug custody paperwork to the medical/legal system; the flight surgeon's protocol book as its own deliverable.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-204, v1.0 — CURRENT — PASS AS AN INVENTORY/DESIGN DOCTRINE; MEDICAL VALIDATION SEPARATE FROM PHYSICS, seated law, master v2.1 (numerical print order). Built 24 August 2026, fifth of the 20-chapter 'ok i will do last nights now you do the next 20' shift (the author's order, 24 August 2026, seated in sitting 607). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597 and 607): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 204